

# Angqi Li

liangqi1@msu.edu | 517-580-1321 | 431 Glenmoor Rd, East Lansing, MI

## Education

- Michigan State University**, Ph.D. in Computational Mathematics, and Science Engineering Jan 2023 – Present
- Michigan State University**, B.S. in Chemical Engineering Sept 2018 – May 2022
- **Minor:** Computer Science & Mathematics

## Experiences

- Research Assistant**, Michigan State University Jan 2023 – Present
- Applied machine learning and deep learning techniques to analyze neural signals and medical images, improving data-driven insights in neuroscience research.
  - Utilized supervised and unsupervised learning methods (e.g., CNNs, RNNs, transformers, and clustering algorithms) to uncover patterns in brain activity and medical imaging.
  - Collaborated with neuroscientists, psychologists, and engineers to develop interdisciplinary solutions for brain signal decoding.
- Laboratory Assistant**, Shanghai Jiaotong University – Shanghai, China Apr 2021 – Aug 2021
- Assisted 3 researchers in analyzing data for 4 projects, debugging prototypes.
  - Conducted 2 independent experiments, improving tissue clearing (CUBIC) efficiency by 10%.
  - Played a key role in developing new non-toxic, organic transparency methods as part of the core research team.

## Projects

- DiffuseClean-** A Diffusion Model for EEG Artifact Removal
- Built a diffusion model to remove ocular artifacts, improving EEG signal quality.
  - Achieved 10% lower RRMSE and 5% higher Cosine Similarity, outperforming traditional methods.
- Fine-Tuning a Large Language Model**
- Trained and fine-tuned a Large Language Model (LLM) for domain-specific applications using PyTorch and Hugging Face Transformers.
  - Deploying the model for chatbots and automated summarization.

## Activities

- Organizer and Co-host**, Data-driven Understanding of Meditation and Consciousness (DUNES) Webinar Series May 2024 - Present
- Expanded global collaboration by organizing and co-hosting a webinar series on Zoom.
  - Engaged 300 researchers and students from 5 countries, fostering interdisciplinary dialogue.

## Skills

- **Programming Languages:** Python, C++ , C, R, MATLAB, SQL, JavaScript
- **Machine Learning:** Data Preparation, Feature Engineering, Model Optimization
- **Data Science Tools:** Jupyter Notebooks, TensorFlow, PyTorch, Scikit-learn

## Publications and Conferences

- **Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering** ICASSP 2024  
Siddhant Gautam, *Angqi Li*, Saiprasad Ravishankar

10.1109/ICASSP48485.2024.10446528

- **Understanding Longitudinal Effects of Mantra Meditation and Breath-focused Meditation using EEG** Neuroscience 2024  
Chicago  
*Angqi Li*, Pratham Pradhan, Krishna Ika, Mengsen Zhang, Barry H. Cohen, Saiprasad Ravishankar
- **Measuring the Effectiveness of Mantra-based Meditation using EEG Data Analysis** Neuroscience 2023  
Washington, D.C.  
*Angqi Li*, Pratham Pradhan, Annie Wozniak, Krishna Ika, Barry H. Cohen, Saiprasad Ravishankar
- **Patient-adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering** Neurips 2023  
Siddhant Gautam, *Angqi Li*, Evan Bell, Saiprasad Ravishankar